



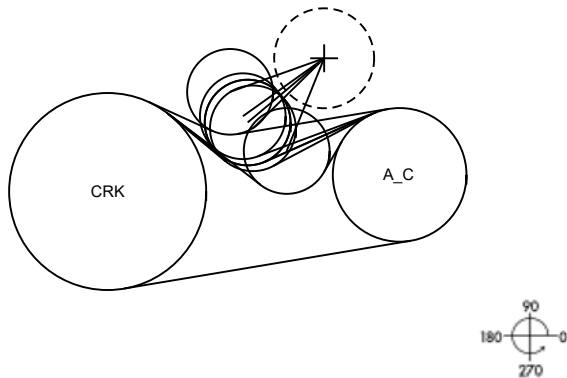
**AG0868 BMC-Otomotif- 6Cyln.
Secondary AC (SD7H15) Drive
Tensioner; E9843/22.5Nm@28°
Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**

Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

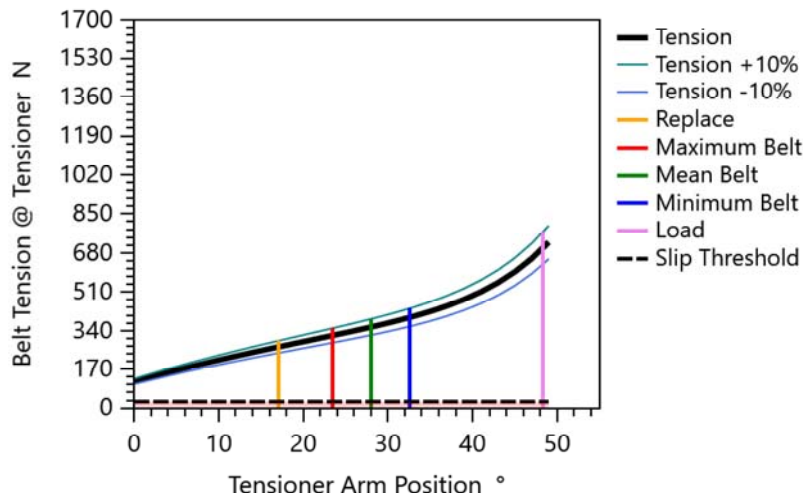
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length mm (ISO 9981)	1020

Tensioner Data

Type	Automatic
Design Tension N	356 (80 lbf)



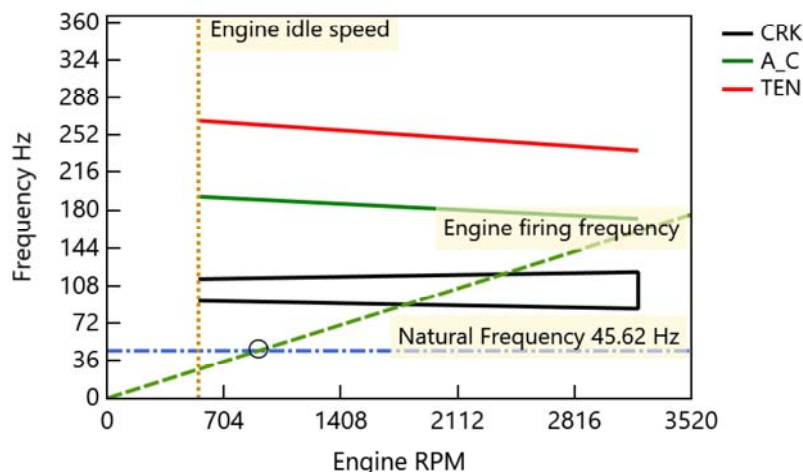
Belt Tension Control



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	24355

Natural Frequency Map



Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
CRK	1400	1000	724	997.2	354 / 228
A_C	1400	1000	362	1078.1	194 / 195
TEN	550	1000	356	373.6	84 / 63

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

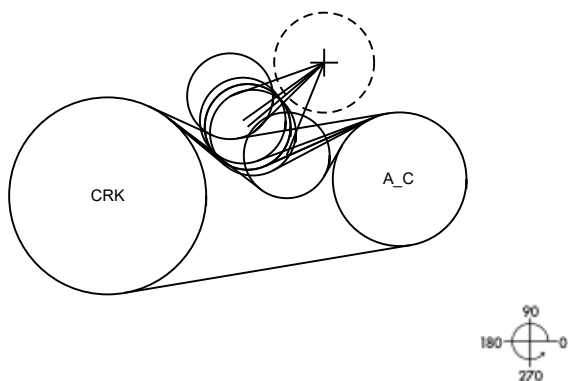
Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->CRK		4.11	100	
CRK->A_C	TEN (1.20)	INCAPABLE		100



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Layout Data mm

	X	Y	Flat	Pitch	Effective	DOB
CRK	0.00	0.00		177.41	175.01	176.00
A_C	263.00	15.00		120.41	118.01	119.00
TEN	126.06	62.15	75.00	77.20	79.60	



Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.70
Belt Length Tolerance (+/-) mm	5.00

Tensioner Data

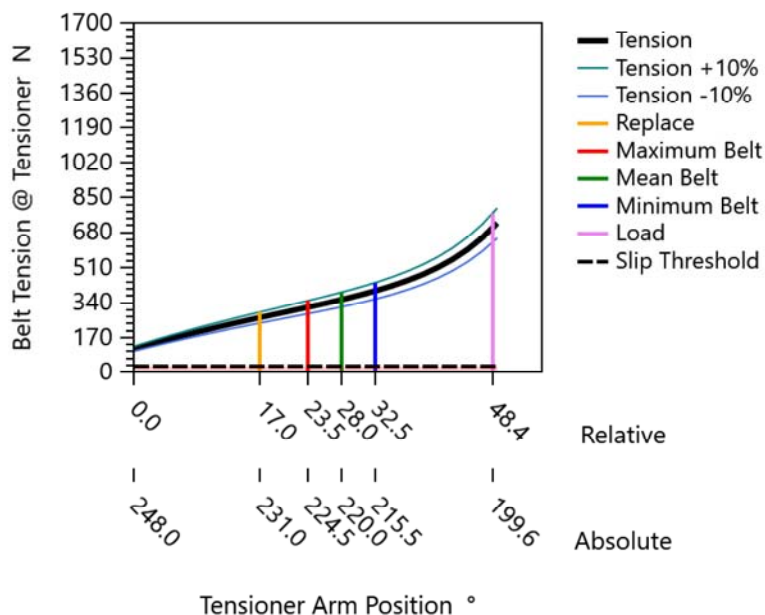
Type	Automatic
Design Tension N	356 (80 lbf)
Pivot Point {X, Y Coordinates} mm	195.00 , 120.00
Arm Length mm	90.0
Spring Mean Load Nm	22.57
Spring Pre-Load Nm	8.56
Spring Rate Nm/deg	0.501

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
CRK	261.9	228.2	1.000
A_C	105.9	194.5	1.473
TEN	59.5	62.7	2.298

Belt Tension Control





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8PK1020HD Polyester; 27Dec2022

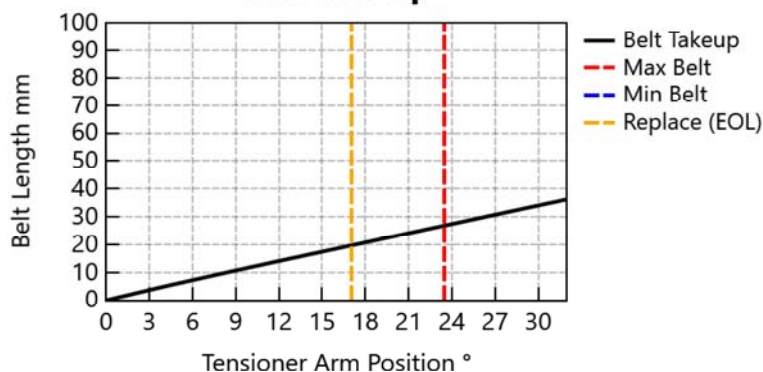
Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/°

1.108

Position Description	Belt-Drive Length					
	Free Arm	Replace	Max	Mean	Min	Load
Arm Position °	248.0	231.0	224.5	220.0	215.5	199.6
Idler X Coordinate mm	161.3	138.3	130.8	126.1	121.7	110.2
Idler Y Coordinate mm	36.6	50.1	56.9	62.1	67.8	89.8
Hubload to Arm Angle °	33.2	36.0	39.6	42.7	46.3	62.9
Hubload Direction °	101.2	87.0	84.1	82.7	81.8	82.5
Hubload N	173.9	322.8	354.3	370.1	381.9	409.0
Tension N	115.7	268.2	318.8	355.6	397.9	702.1
Idler Wrap Angle °	97.4	74.0	67.5	62.7	57.4	33.9
Effective Drive Length mm	1051.5	1031.9	1024.7	1019.7	1014.7	999.1
Required Eff. Belt Length mm	1051.8	1032.2	1025.0	1020.0	1015.0	999.4

Belt Take-up

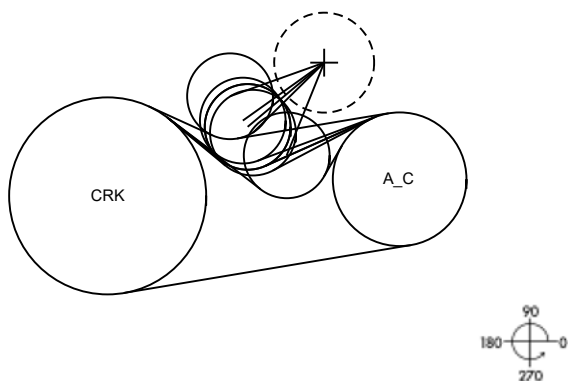




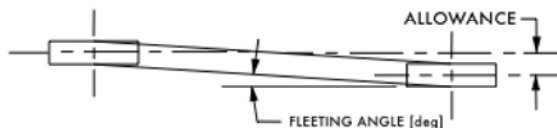
**AG0868 BMC-Otomotif- 6Cyln.
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Tensioner; E9843/22.5Nm@28°
Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**

Alignment Data °

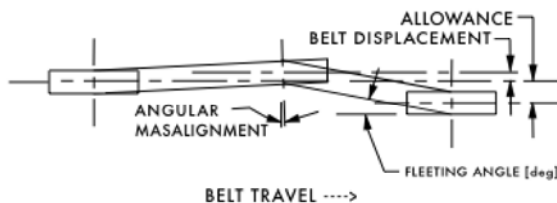
Fleeting Angle	0.90
TEN Angular Misalignment	1.20



(GROOVED - GROOVED) PULLEYS



(GROOVED - FLAT - GROOVED) PULLEYS



RESULTS

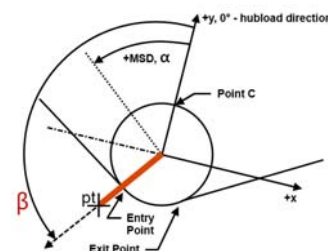
Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->CRK		4.11	100	
CRK->A_C	TEN (1.20)	INCAPABLE		100

Hubload to Arm Angle & MSD [°]



Hubload-to-arm angle is the angle between hubload and arm where arm direction is from pulley center to pivot.

β = Hubload-to-arm angle

pt = Tensioner pivot point

Backside Pulley Alignment Sensitivity

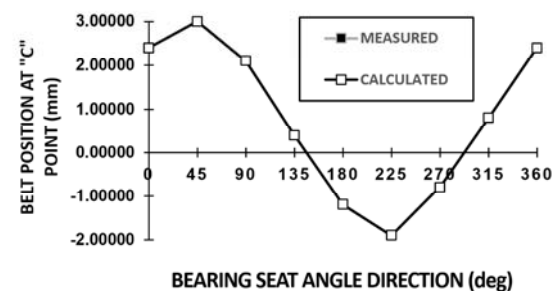
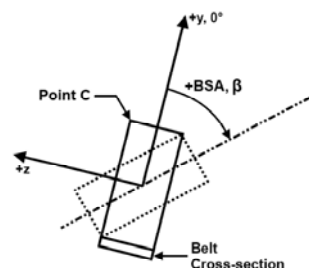
Backside Pulley	Angular Misalignment	Most Sensitive Direction	Resulting Fleeting Angle	Suggestions
3: TEN	1.200	74.36	0.91	AB C E

Suggestion List for Backside Pulley Alignment Sensitivity

- A] Increase EXIT span length
- B] Decrease ENTRANCE span length
- C] Increase both ENTRANCE / EXIT span lengths
- D] Increase wrap angle
- E] Decrease wrap angle

Note: Most Sensitive Direction for Grooved pulleys is parallel to the belt entry span.

Direction of Bearing Seat Angle [°]





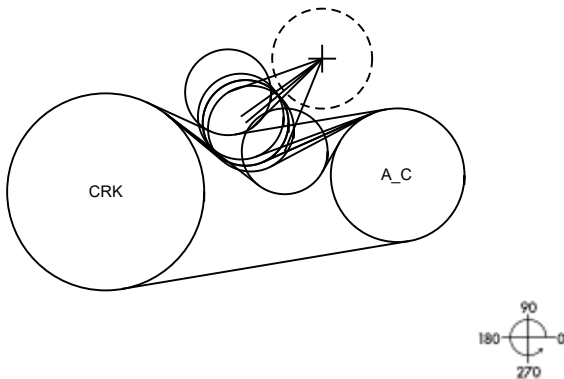
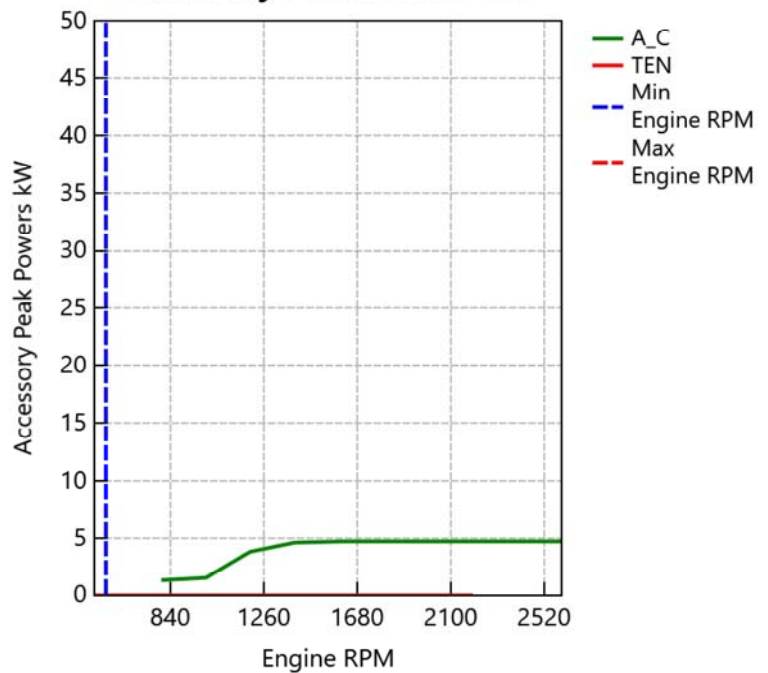
User: Abdurrahman Kol

Design: AG0868_BMC-Power AC-SD7H15_Drive_8PK1020HD_E9843-22.5Nm_27Dec2022

**AG0868 BMC-Otomotif- 6Cyln.
Secondary AC (SD7H15) Drive
Tensioner; E9843/22.5Nm@28°
Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**

Design Tension N	356 (80 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000

Accessory Peak Powers kW



Accessory Data

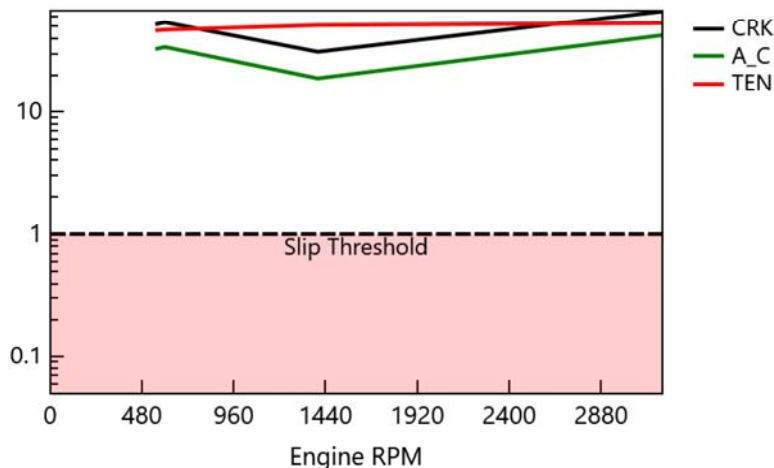
	File Name - if used	Moment of Inertia kg m ²
A_C	SD7H15-AC.cmp	0.0034
TEN	MX14D_MY27_ALT-24V-150Amp-EU-F...	0.0009

RESULTS

Belt Slip Sensitivity

Pulley	Most Critical Load Condition					Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %	A_C			N	°
CRK	1000	100					
A_C	0	100					
TEN	1000	100					

Belt Slip Safety Factor





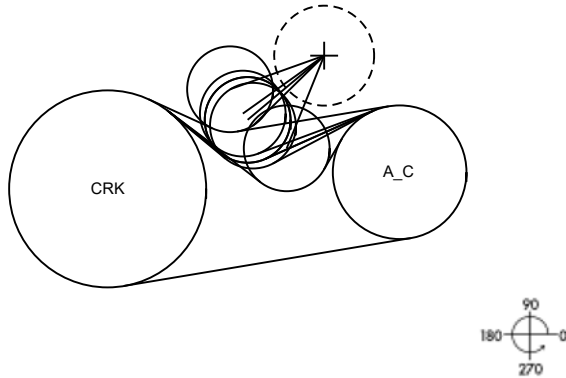
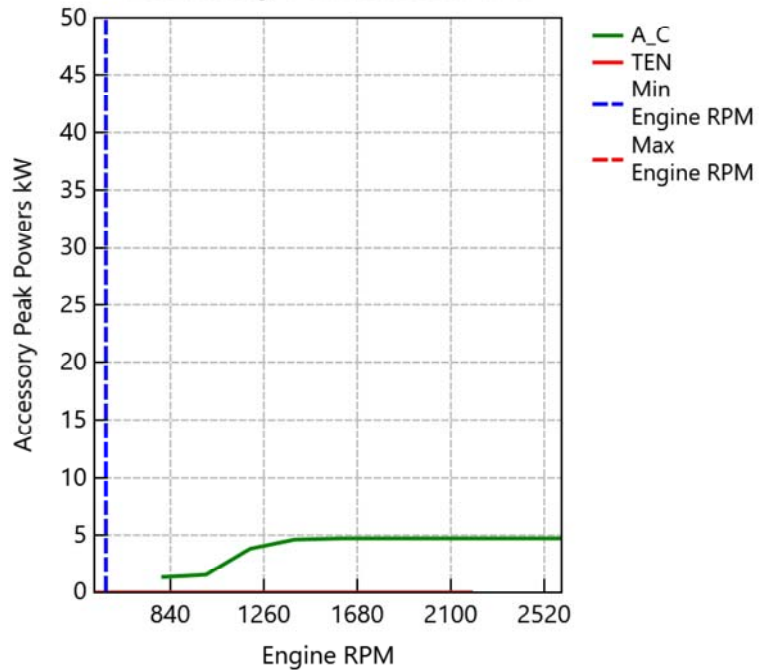
User: Abdurrahman Kol

Design: AG0868_BMC-Power AC-SD7H15_Drive 8PK1020HD_E9843-22.5Nm_27Dec2022

**AG0868 BMC-Otomotif- 6Cyln.
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8PK1020HD Polyester; 27Dec2022**

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Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000

Accessory Peak Powers kW



RESULTS

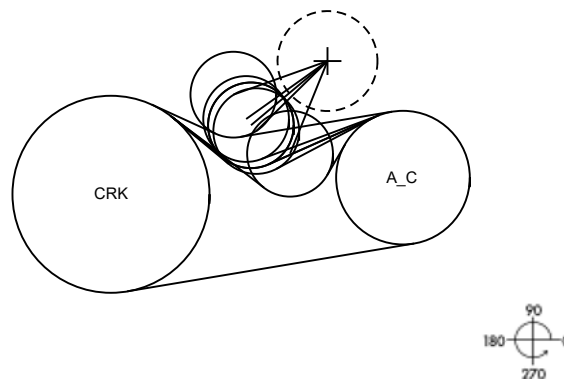
Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °	Shear Stress N/mm-rib @ RPM
CRK	1400	1000	724	997.2	354 / 228	0.134 @ 1400
A_C	1400	1000	362	1078.1	194 / 195	0.231 @ 1400
TEN	550	1000	356	373.6	84 / 63	N/A

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)



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Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**



Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	A_C	TEN
27.00	800	1	90	1.31	1.30	0.01
10.00	1000	1	90	1.51	1.50	0.01
13.00	1250	1	90	4.01	4.00	0.01
18.00	1500	1	90	4.61	4.60	0.01
19.00	1750	1	90	4.71	4.70	0.01
13.00	2000	1	90	4.71	4.70	0.01

Mean Hubloads N for DC 95%

%	Eng RPM	CRK	A_C	TEN
27.00	800	813	882	371
10.00	1000	801	868	371
13.00	1250	975	1050	371
18.00	1500	961	1035	370
19.00	1750	921	994	370
13.00	2000	887	958	370

Mean Tensions N for DC 95%

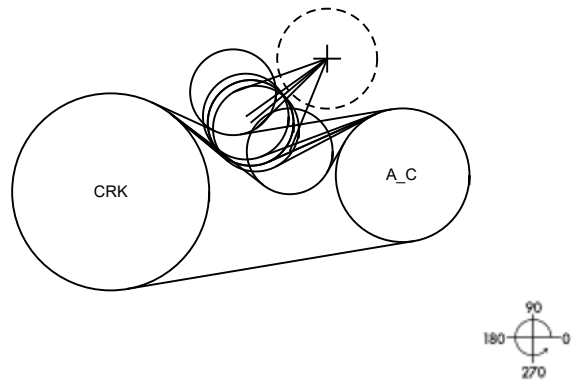
%	Eng RPM	CRK	A_C	TEN
27.00	800	532	357	356
10.00	1000	518	357	356
13.00	1250	701	356	356
18.00	1500	686	356	356
19.00	1750	645	356	356
13.00	2000	609	356	356



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Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**

Load Conditions [kW]	
Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016	
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester

Tensioner Data	
Type	Automatic
Design Tension N	356 (80 lbf)





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Secondary AC (SD7H15) Drive
Tensioner; E9843/22.5Nm@28°
Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**

RESULTS

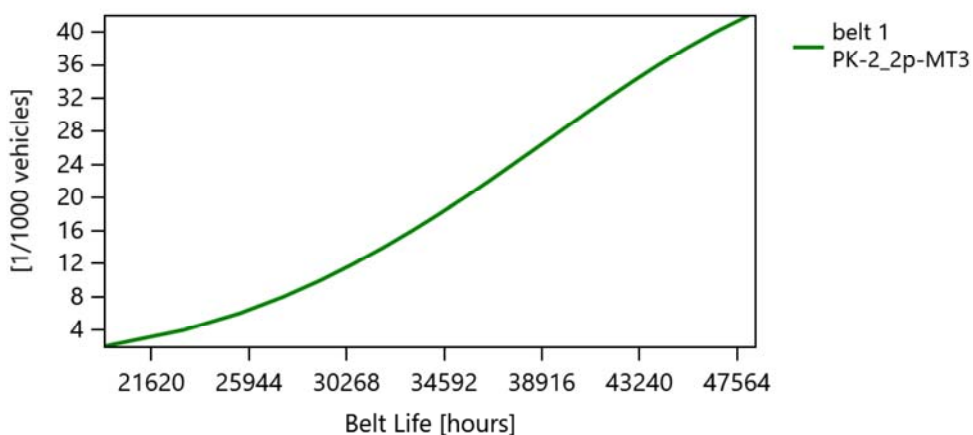
Pulley Contributions to Belt Rib Fatigue %

	CRK	A_C	TEN
PK-2_2p-MT3	2.7	25.0	72.4

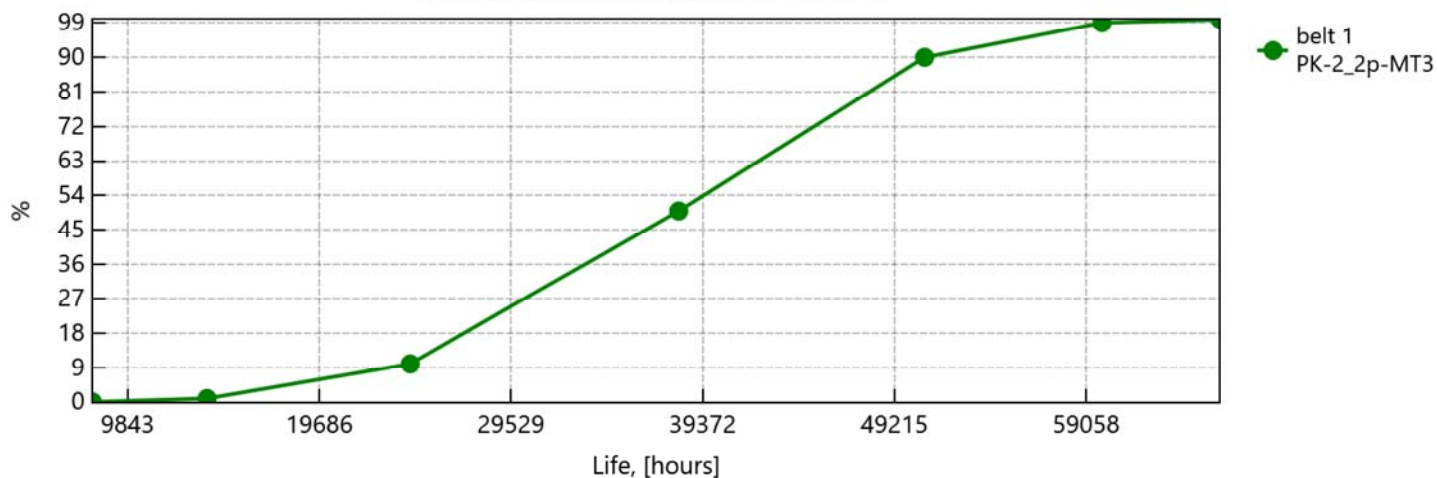
Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	24355

Belt Incidents per Thousand Vehicles



Weibull Distribution for DC95%



Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	A_C	TEN
27.00	800	1	90	1.31	1.30	0.01
10.00	1000	1	90	1.51	1.50	0.01
13.00	1250	1	90	4.01	4.00	0.01
18.00	1500	1	90	4.61	4.60	0.01
19.00	1750	1	90	4.71	4.70	0.01
13.00	2000	1	90	4.71	4.70	0.01

Load Contribution to Belt Rib Fatigue for DC 95%

%	Vblt m/s	CRK Tt N	CRK Ts N	% PK-2_2p-MT3
27.00	7.4	532	356	16
10.00	9.3	518	356	7
13.00	11.6	701	356	13
18.00	13.9	686	356	21
19.00	16.3	645	356	25
13.00	18.6	609	356	18



**AG0868 BMC-Otomotif- 6Cyln.
Secondary AC (SD7H15) Drive
Tensioner; E9843/22.5Nm@28°
Tensioner Position 195/120
8PK1020HD Polyester; 27Dec2022**

Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester

Tensioner Data

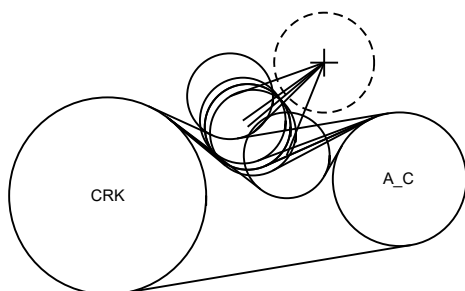
Type	Automatic
Design Tension N	356 (80 lbf)
Inertia of the Tensioner Arm Kg-m ²	0.0009
Tensioner Pulley Mass kg	0.80

Engine Data

# of Cylinders	6
Crankshaft Moment of Inertia kg m ²	0.5000

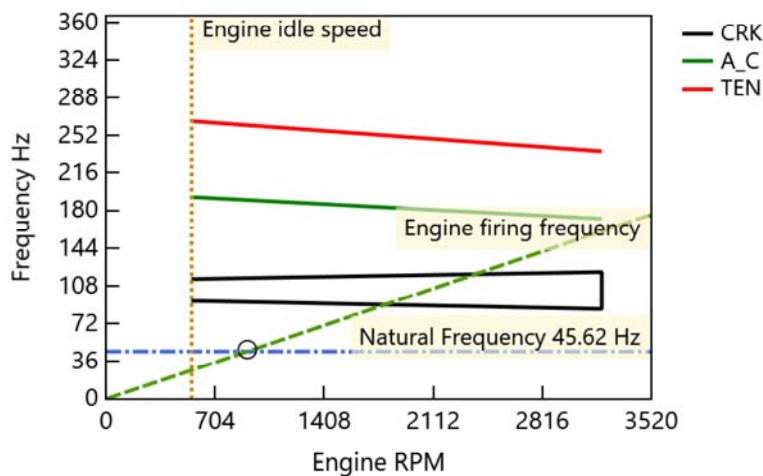
Accessory Data

	File Name - if used	Moment of Inertia kg m ²
A_C	SD7H15-AC.cmp	0.0034
TEN	MX14D_MY27_ALT-24V-150Amp-EU-...	0.0009



RESULTS

Natural Frequency Map



Note: Only low frequency spans are shown.
System Resonance (Mode 1: 45.55)

Note: If Engine Idle is near system resonant RPM,
then accessory drive system may have
excessive vibration.
If system resonance is within a span
frequency window, then span vibration
may have excessive amplitude.



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Drive Notes

Date	Note
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